Glance ML Internship Assignment

Multimodal Fashion and Context Retrieval

Project Overview

The assignment asks for an image retrieval system that understands both fashion attributes and scene context. A plain CLIP baseline is a useful starting point, but it often fails on compositional prompts such as 'red tie and white shirt' because the whole query is compressed into one vector. I built a two-stage retrieve-and-rerank system that keeps the fast zero-shot benefits of CLIP while adding garment-level checks for color, category, and person association.

1. Approaches and Tradeoffs

Approach	Strength	Tradeoff	Best use
Plain CLIP	Very simple; fast ANN search; strong zero-shot baseline.	Weak at binding colors to garments; scene and attributes compete in one vector.	Broad vibe search where strict garment/color comp
Supervised detector + classifiers	Precise when taxonomy is fixed and labels are available.	Limited zero-shot vocabulary; new garments need new labels or training data.	Catalog search with a controlled product taxonomy
Open-vocabulary retrieve + rerank	Keeps zero-shot recall while checking region-level fashion constraints.	More moving parts and slower than a single CLIP lookup.	Fashion queries with colors, garments, context, an

2. Chosen Approach

Indexer

The indexer processes each image offline. First, it stores a full-image embedding from Marqo FashionCLIP (hf-hub:Marqo/marqo-fashionCLIP). Then Grounding DINO detects garments and person boxes. For every garment crop, the pipeline stores a local CLIP embedding and a dominant RGB color extracted with K-means. Garments are assigned to the most-overlapping person box, so later retrieval can check whether multiple requested garments belong to the same person. ChromaDB stores one vector document per image, with region metadata serialized alongside the full-image embedding.

Retriever

At query time, Llama 3.3 70B through Groq parses the text into garment/color constraints and an optional setting phrase. The raw query is embedded with FashionCLIP and ChromaDB returns the top 300 candidates. These candidates are reranked with a weighted combination of global similarity, garment/color attribute score, and setting similarity.

How it handles fashion queries

For garment matching, the reranker compares query label embeddings against stored region embeddings rather than relying only on detector labels. It expands synonyms, so 'raincoat' can match regions that look like coats, jackets, or anoraks. Color phrases are normalized before RGB distance checks; for example, 'bright yellow' becomes 'yellow' and 'navy blue' becomes 'navy'. For multi-garment prompts, the score is computed per person instance. This prevents a red tie on one person and a white shirt on another from being treated as a full compositional match.

3. Codebase Link

Repository: https://github.com/tanishq170205/Fashionpedia.git

The repository separates indexing, retrieval, evaluation, and the optional FastAPI demo app. The main files are indexer/main.py for offline indexing, retriever/main.py for command-line search, retriever/reranker.py for the compositional logic, and eval/run_eval.py for the five benchmark queries.

4. Limitations and Future Work

Dataset coverage

Fashionpedia val_test2020 is mostly runway, editorial, and fashion-event imagery. Office interiors and park-bench scenes are not common, so context-heavy queries are partly limited by dataset coverage rather than only retrieval logic. The code still handles the query structure, but no retrieval method can return many correct office or park images if they are not present in the corpus.

Adding locations and weather

The practical extension is metadata enrichment. If GPS and timestamps are available, index city, region, season, temperature, and weather through EXIF and a historical weather API. If metadata is missing, add image-based classifiers for coarse location type and weather class. The query parser can then emit optional fields such as location='Tokyo' or weather='rainy', and ChromaDB can apply these as metadata filters before ANN retrieval. This is preferable to hoping CLIP can infer real geography or weather from pixels alone.

Improving precision

Precision can be improved in four directions. First, fine-tune the region encoder on Fashionpedia train masks and attributes so garment concepts are better separated. Second, replace hand-set fusion weights with a small learned reranker trained from relevance judgments. Third, add pattern and material recognition for striped, plaid, leather, denim, or silk prompts. Fourth, replace bounding-box person association with pose or segmentation for crowded and occluded scenes.

Scalability

The architecture scales because only stage 1 touches the full corpus. With one million images, HNSW still returns a fixed candidate set and stage 2 only reranks those candidates. The main production change would be storing garment regions in a separate region-level index instead of embedding all region vectors inside image metadata.